



SIMATS ENGINEERING
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CSA0406 - OPERATING SYSTEMS

**Comparative Study of Popular Operating Systems
(Windows, Linux, macOS, Android)**

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Introduction

❑ **What is an Operating System?**

- An Operating System (OS) is system software that manages computer resources and acts as a bridge between the user and the hardware.

❑ **Main Functions :**

- ✓ Manages CPU and memory
- ✓ Handles files and storage
- ✓ Controls hardware devices
- ✓ Provides security
- ✓ Runs applications



Popular Operating Systems

□ **Windows**

- Developed by Microsoft
- Popular for PCs and laptops
- Easy to use

□ **Linux**

- Open-source operating system
- Secure and customizable
- Used in servers

□ **macOS**

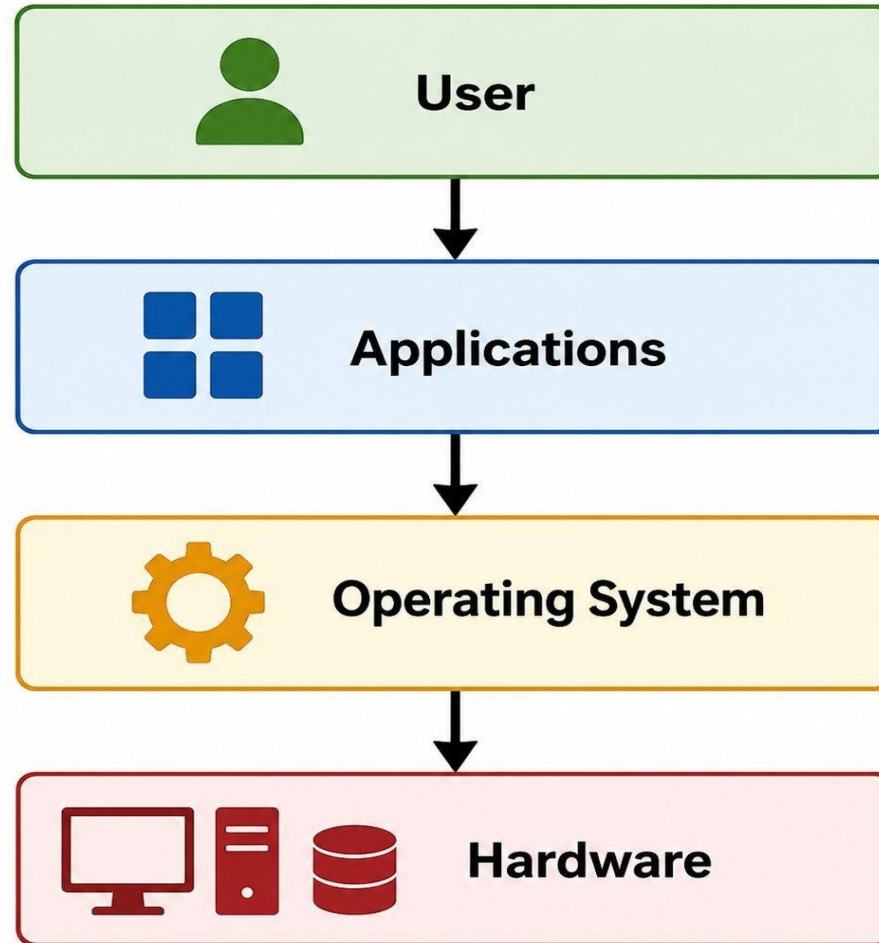
- Developed by Apple
- Used on Mac computers
- Fast and secure

□ **Android**

- Developed by Google
- Most popular mobile OS
- Used on smartphones and tablets



Diagram





Advantages and Disadvantages

Operating System	Advantages	Disadvantages
Windows	Easy to use, supports many applications	Paid license, higher risk of malware
Linux	Free, secure, highly customizable	Steeper learning curve
macOS	Stable, secure, fast performance	Expensive hardware
Android	Flexible customization, vast selection of apps	Security varies depending on the device



Performance Comparison





THANK
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